

# Interpretable Gene Programs or Foundation Models?

Comparing Sparse Dictionary Learning and scGPT for Single-Cell Classification

## Abstract

Single-cell foundation models such as scGPT learn representations from large-scale gene-expression data, but the structure captured by these representations can be difficult to interpret. We propose a controlled comparison between pretrained scGPT and sparse dictionary learning, an interpretable representation-learning method that expresses cells as sparse combinations of learned gene programs. Using a public single-cell RNA-sequencing dataset, both representations will be evaluated using the same regularized multinomial logistic regression classifier, allowing us to compare their usefulness for downstream cell-type classification. A classifier using the processed gene-expression features directly will provide an additional baseline.

Dictionary learning approximates the expression matrix as  $X \approx ZD$ , where  $D$  contains learned gene programs and  $Z$  contains their activities within individual cells. Sparsity is encouraged through the objective

$$\min_{Z,D} \frac{1}{2} \|X - ZD\|_F^2 + \lambda \|Z\|_1,$$

providing a theoretical connection to sparse representation learning and  $L_1$ -regularized estimation. Empirically, we will compare representations using Macro-F1, balanced accuracy, and predictive log loss, while examining the highest-weight genes and cell-type activation patterns associated with learned dictionary atoms. The project asks whether pretrained scGPT provides downstream predictive information beyond a sparse, directly interpretable representation learned from the target data. More broadly, we aim to characterize the tradeoff between predictive performance and interpretability in single-cell representation learning.